
CODE SANGAM 2026
MNNIT
COMPUTER CODING CLUB
Droidrush

✦ CODE SANGAM 2026 // DROIDRUSH ✦

RULES & REGULATIONS

All participants are required to strictly follow the guidelines and rules outlined below.

I. TEAM & ELIGIBILITY

- **Team Size:** Maximum of 3 members per team. You can also participate individually.
- **Eligibility:** Team members can be from any branch, but must be from the same academic year.
- **Core Objective:** Build a working Android or iOS application for any **one** problem statement before the deadline.

II. PLATFORM & ABSTRACT

- **Platform Freedom:** Native (Kotlin/ Compose, Swift) or Hybrid (Flutter, React Native) platforms can be used. PWAs are not allowed.
- **Abstract Submission:** The abstract template is provided on the official website—**strictly adhere to it**.
- **Mentorship Support: No mentor will be allotted**; however, you can reach out to any Computer Club (CC) member or coordinator for queries.

III. REPOSITORY & COMMITS

- **Repository Genesis:** The GitHub repository must be created *only after* the development period officially begins.
- **Commit Freeze:** No commits are allowed once the development period ends. Commit timestamps will be audited.
- **Cumulative Scoring:** This is a cumulative scores event; you need to perform consistently across all rounds to win.

IV. INTEGRITY & CONDUCT

- **Zero Plagiarism:** ANY form of code plagiarism will not be tolerated and will lead to immediate disqualification.
- **Judges' Decision:** The decision of the judges will be final and binding on all participants.
- **Doubts & Help:** For any discrepancies or help at any stage, contact seniors or reach out to CC members.

SUBMISSION & VIVA DEFENSE

Ensure your repository has a clean commit history and a descriptive README. Teams must be prepared to defend their technical architecture and implementation in the live viva.


Easy

01

EYE OF AGAMOTTO RUNIC SCRIBE

POINT, READ, AND TRANSMUTE GLYPHS OFFLINE

The mortal realm is inundated with physical texts, signs, and ledgers lost in ink. Channel the talisman's optical sensor to transcribe reality into digital thought entirely on-device, extracting text instantly without transmitting a single syllable to external cloud servers.

OBJECTIVE

Build a high-speed Android text grabber using CameraX and on-device ML Kit Text Recognition v2 that detects live bounding boxes, copies tapped text blocks, and logs scan history offline.

FOUNDATIONAL SCOPE (PASS TIER)

- **CameraX Optical Feed:** Live preview + on-device ML Kit Text Recognition v2 with reactive bounding boxes.
- **Tap-to-Transcribe:** Tap any detected text block to copy, freeze-frame capture, and share-as-text.
- **Strict Offline Realm:** Works 100% without internet, including emulator verification with bundled sample image.
- **Room Ledger:** Persistent scan history (thumbnail, extracted text, timestamp) with search and deletion.

ASCENDED MASTERY (VICTORY TIER)

- **Script Auto-Detect:** Latin, Devanagari, and more with on-device ML Kit Translation integration.
- **Smart Parse Modes:** Tappable phone numbers, URLs, and emails; receipt parser extracting total and date.
- **Archival Export:** Export history to CSV/PDF, flashlight toggle, and batch multi-page scan merge.



DELIVERABLES

- Public GitHub Repository with clean commit history
- README: Architecture (CameraX vs ML Kit pipeline) + 3 failure modes
- Tool disclosure & live offline test proofs

TECHNICAL FREEDOM

Construct using Native Android (Kotlin / Jetpack Compose) or Flutter. Fully on-device ML inference; zero cloud dependencies. Bonus marks for offline translation speed and clean Room indexing.

02

ASTROLABE OF THE VISHANTI

YOUR TALISMAN AS AN ONBOARD ARCANE INSTRUMENT

Every handheld device harbors dormant hardware senses tuned to gravity, planetary magnetism, and spatial velocity. Awaken these latent instruments to construct a real-time diagnostic laboratory on-device, charting physical orientation and magnetic anomalies.

OBJECTIVE

Expose onboard hardware sensors via SensorManager as real-time responsive diagnostic instruments featuring live telemetry graphs, a smoothed bubble level, a rotation compass, and a calibrated magnetometer.

FOUNDATIONAL SCOPE (PASS TIER)

- **Sensor Enumeration:** Query SensorManager and graph live streams (accel, gyro, mag, light, proximity) with adjustable sampling rate.
- **Planar Calibration:** Smoothed bubble level + protractor from accelerometer and compass from rotation-vector sensor.
- **Session Telemetry:** Record sensor sessions to CSV and export/share with live telemetry readouts.
- **Graceful Degradation:** Safe null-sensor handling and simulation fallbacks on emulators.

ASCENDED MASTERY (VICTORY TIER)

- **Magnetic Anomaly Detector:** Magnetometer metal detector with threshold calibration and haptic/audio alerts.
- **Sensor-Fusion Kinematics:** Orientation (pitch/roll/yaw), free-fall & shake detection, and step-counter sensor integration.
- **Chrono-Replay:** Replay recorded sensor sessions as animated graphs, plus barometer relative altitude.

DELIVERABLES

- GitHub Repository with incremental commit progression
- README: Sensor-fusion mathematics, coordinate frames & failure modes
- Tool disclosure & emulator/device sensor logs

TECHNICAL FREEDOM

Construct using Native Android (Kotlin/Compose) or Flutter. Strictly local hardware sensor polling; zero external telemetry. Bonus marks for battery-efficient sampling and low-jitter smoothing algorithms.

03
THE CHRONOS SEAL

DISCIPLINES THAT INSCRIBE THE ASTRAL PLANE

Daily disciplines and mystic oaths dissolve without immutable consistency. Construct an offline habit ledger bound directly to the user's primary desktop through responsive home-screen widgets, tracking unbroken streaks without whispering a byte to the cloud.

OBJECTIVE

Deliver a privacy-centric offline habit tracker featuring interactive home-screen App Widgets for instant 1-tap completions, persistent streak telemetry, and localized alarm schedules.

FOUNDATIONAL SCOPE (PASS TIER)

- **Room Habit Core:** Create habits (name, icon, color, schedule) with streak tracking and completion rates in Room.
- **Home-Screen Widget:** Interactive App Widget listing today's habits with tap-to-complete updating instantly.
- **Local Temporal Wake:** Daily reminder alerts scheduled via WorkManager / AlarmManager with zero network calls.
- **Sovereign Vault:** Fully offline operation with local JSON backup and restore.

ASCENDED MASTERY (VICTORY TIER)

- **Chronos Heatmap:** Calendar matrix, longest-streak statistics, and a 'freeze day' shield protecting streaks.
- **Dynamic Canvas:** Multiple widget dimensions (streak ring, week grid), Material You dynamic theming.
- **Granular Rituals:** Partial-completion tracking (e.g. 8 water units), custom notification times, progress cards.



DELIVERABLES

- GitHub Repository with complete source code
- README: App Widget state synchronization & WorkManager lifecycle
- Tool disclosure & local data schema documentation

TECHNICAL FREEDOM

Construct using Native Android (Kotlin / Jetpack Glance / Compose) or Flutter. Zero cloud databases. Bonus marks for seamless App Widget performance, zero-latency Room sync, and elegant widget layouts.


Medium

04
THE WARD OF SILENCE

TAMING THE ETHEREAL NOTIFICATION INUNDATION

An endless barrage of external broadcasts fractures mortal concentration. Establish an ambient guardian via background notification listeners that intercepts, categorizes, mutes, or digests incoming dispatches according to strict deterministic laws.

OBJECTIVE

Engineer an offline notification management system using NotificationListenerService that captures, filters, groups, and summarizes alerts with complete privacy and zero spam injection.

FOUNDATIONAL SCOPE (PASS TIER)

- **Listener Sentinel:** NotificationListenerService with clear permission flow; logs app, title, text, timestamp to Room with sensitive fields redacted.
- **Deterministic Law:** Rule engine: WHEN app is X AND text contains Y THEN mute / mark important / route to channel (first-match evaluation).
- **Filterable Chronicle:** Searchable history timeline grouped by app with mark-read and clear-all capabilities.
- **Privacy Shield:** Purely read-only observer; strictly zero auto-reply or unauthorized message generation.

ASCENDED MASTERY (VICTORY TIER)

- **Astral Digest:** Hourly/daily digest bundling low-priority notifications, plus per-app DND quiet hours.
- **Contextual Classification:** On-device keyword tagging (OTP, promo, social) and 'snooze until' re-posting via WorkManager.
- **Telemetry Vault:** Activity metrics dashboard (noisiest apps, hourly rate), CSV export, encrypted SQLCipher storage.



DELIVERABLES

- Public GitHub Repository with documented commits
- README: NotificationListenerService security model & 3 edge-case failures
- Tool disclosure & privacy consent documentation

TECHNICAL FREEDOM

Construct using Native Android (Kotlin). Strictly local rule matching; zero external analytics or network dispatch. Bonus marks for bulletproof sensitive data scrubbing and low battery drain.

05
ELDRITCH AUTOMATA

DETERMINISTIC LAWS OF THE HANDHELD CHAMBER

When physical contexts alter, the talisman must enact defensive countermeasures without manual intervention. Build an explicit background trigger-action engine that senses environmental states, evaluates cascading logic gates, and logs execution with rollback.

OBJECTIVE

Build a deterministic device automation engine with a visual rule-builder UI that detects hardware states (Wi-Fi, charging, time) and executes system actions with undo and dry-run preview.

FOUNDATIONAL SCOPE (PASS TIER)

- **Visual Rule Forge:** Triggers (time, Wi-Fi SSID, headphones, battery, charging) and actions (ringer/DND, launch app, toggle Wi-Fi/BT, post alert).
- **Daemon Evaluation:** Reliable background processing via WorkManager/foreground service + BroadcastReceivers (first-match execution).
- **Reboot Invariant:** Survives device reboot (BOOT_COMPLETED); features dry-run preview before committing rules.
- **Undoable Journal:** Run log with timestamp, trigger, and executed action, plus one-tap rollback where reversible.

ASCENDED MASTERY (VICTORY TIER)

- **Advanced Gates:** Geofencing & calendar triggers, AND/OR/NOT combinators, rule priority & recursion loop prevention.
- **Tactical Actions:** Auto-SMS reply while driving (explicit consent), NFC tag triggers, and Quick-Settings tile toggles.
- **Crash-Safe Journal:** WAL-backed run journal, per-rule cooldown limits, and JSON rule-set export/import.



DELIVERABLES

- GitHub Repository with clean architectural separation
- README: State machine diagrams, receiver security & loop prevention
- Tool disclosure & verified automation logs

TECHNICAL FREEDOM

Construct using Native Android (Kotlin). Explicit rules only; zero black-box LLM classifiers. Bonus marks for low-power geofencing, crash-safe transaction logs, and clean broadcast receivers.

06
THE MIRROR ARCHIVE

AUTONOMOUS CURATION OF THE VISUAL MEMORIES

Gigabytes of visual memories lie in disarray within local storage. Deploy an autonomous background librarian that indexes images, infers semantic concepts using on-device vision models, identifies near-identical duplicates, and keeps every photograph locked to local storage.

OBJECTIVE

Deliver an on-device gallery organizer that scans MediaStore images via WorkManager, tags them with ML Kit Image Labeling, auto-sorts albums, and detects near-duplicates using image hashing.

FOUNDATIONAL SCOPE (PASS TIER)

- **Background Ingestion:** Asynchronous MediaStore indexing via WorkManager with real-time progress indicators.
- **Semantic Vision:** On-device ML Kit Image Labeling automatically tagging detected objects and scenes.
- **Auto-Categorized Albums:** Dynamic albums (receipts, screenshots, documents, food, people) with tag-search UI.
- **Perceptual Hashing:** Near-duplicate detection via dHash/pHash with side-by-side comparison and purge flow.

ASCENDED MASTERY (VICTORY TIER)

- **Entity Clustering:** On-device face grouping (ML Kit Face Detection) into person clusters, plus burst 'best shot' selector.
- **Runic Search:** OCR sub-mode with ML Kit text search inside photos; storage-saver identifying blurry images.
- **Incremental Sync:** Incremental re-index on new photos, encrypted index database, collage/zip export.



DELIVERABLES

- GitHub Repository with documented commit lineage
- README: Perceptual hashing algorithm analysis & indexing state machine
- Tool disclosure & on-device performance benchmarks

TECHNICAL FREEDOM

Construct using Native Android (Kotlin/Compose) or Flutter. Fully on-device ML Kit models; zero cloud uploads. Bonus marks for rapid perceptual hashing throughput and low memory footprint during indexing.


Lair

07
THE BLACKOUT LEYLINE

HOP-BY-HOP SOS WHEN ALL GATEWAYS SEVER

When catastrophic collapse severs telecommunication towers and Wi-Fi networks go dark, survival demands localized ad-hoc relay. Build a delay-tolerant peer-to-peer mesh chat routing emergency packets hop-by-hop via Bluetooth Low Energy across isolated devices.

OBJECTIVE

Construct a delay-tolerant BLE mesh network where disconnected nodes A and C communicate through intermediate node B using store-and-forward flooding routing in airplane mode with Bluetooth active.

FOUNDATIONAL SCOPE (PASS TIER)

- **BLE Radio Discovery:** Foreground BLE advertising + scanning on shared Service UUID; GATT connect with ~180B chunked writes.
- **Store-and-Forward:** Flood router with message ID deduplication, TTL 3–5, and SQLite outbox with store-forward on reconnect.
- **Multi-Hop SOS:** 1-hop and 2-hop SOS payloads (<280 chars) displaying sender, timestamp, and hop telemetry.
- **Disaster Mode:** Fully operational in airplane mode with Bluetooth enabled; Android mesh node implementation required.

ASCENDED MASTERY (VICTORY TIER)

- **Background Mesh Relay:** Multi-hop background relay with priority queue (SOS alerts prioritized over regular traffic).
- **Payload Reassembly:** Chunked long message reassembly, reconnect loops, SOS beacon mode, and basic cryptographic signing.
- **Transport Escalation:** Dynamic transport switch to Wi-Fi Direct for large binary payload transfer; 4+ node storm stress test.



DELIVERABLES

- GitHub Repository with modular architecture
- README: Binary packet specification, TTL routing proofs, and 3 failure modes
- Tool disclosure & multi-device physical test log

TECHNICAL FREEDOM

Construct using Native Android (Kotlin). Shared portable routing core with native BLE bindings. Bonus marks for battery-conservative GATT connection pooling, zero dropped SOS packets, and graceful reconnect logic.

08
ACOUSTIC DIVINATION

TRANSMISSION BEYOND HUMAN HEARING & MOTION SENSING

Air itself can transmit information through acoustic frequencies imperceptible to mortal ears. Construct an ultrasonic acoustic modem modulating text via Frequency-Shift Keying through device speakers, and detect hand gestures by measuring the Doppler frequency shift.

OBJECTIVE

Build an acoustic communication link using 48 kHz PCM audio loopback transmitting data via 2-FSK in near-ultrasound bands (18.5–19.5 kHz) alongside a Doppler hand-wave motion detector.

FOUNDATIONAL SCOPE (PASS TIER)

- **Acoustic Loopback:** 48 kHz mono 16-bit AudioTrack + AudioRecord loopback with real-time FFT spectrogram rendering to 24 kHz.
- **Audible 2-FSK Modem:** Audible 2-FSK text transmission (5+ chars) phone-to-phone at 1 m with preamble, threshold, and CRC check.
- **Doppler Presence:** Doppler presence detector identifying frequency shifts from physical hand-waves (VOICE_RECOGNITION source).
- **Wire Frame Diagram:** Full frame structure and modulation parameters documented in architecture specifications.

ASCENDED MASTERY (VICTORY TIER)

- **Near-Ultrasound 19 kHz:** Ultrasonic 18.5–19.5 kHz text (10+ chars) at 1 m on 2 different OEMs with calibration sweep log and BER plot.
- **Directional Wave Gestures:** Directional gestures (approach vs recede, >80% accuracy over 10 trials, <300 ms response).
- **Error Resilience:** Reed-Solomon ECC, 1-to-3 acoustic broadcast test, and device compatibility matrix with audible fallback.



DELIVERABLES

- GitHub Repository with DSP pipeline source code
- README: Modulation frame diagrams, FFT windowing math & OEM acoustic variances
- Tool disclosure & physical broadcast test log

TECHNICAL FREEDOM

Construct using Native Android (Kotlin) or C++/NDK audio. Pure local DSP; zero network access. Bonus marks for low Bit Error Rate (BER), noise-rejection filters, and clean spectrogram visualization.

09
THE THIRD EYE SENTINEL

AN AMBIENT ACCESSIBILITY SENTINEL INSIDE ANY REALM

Navigating complex user interfaces requires an omniscient observer that perceives on-screen hierarchies without intruding. Build an accessibility companion that inspects active node trees, highlights focused elements, and synthesizes instant semantic explanations on-device.

OBJECTIVE

Develop an Android AccessibilityService copilot that traverses the active window node hierarchy, projects a non-intrusive focus highlight overlay, and generates contextual summaries on-device.

FOUNDATIONAL SCOPE (PASS TIER)

- **Tree Introspection:** AccessibilityService with window-content retrieval and tree dump with node compression.
- **Focus Overlay:** TYPE_ACCESSIBILITY_OVERLAY drawing high-contrast focus highlights following active UI elements.
- **Deterministic Summarizer:** Rules-only semantic summary with prominent consent disclosure and secure field redaction.
- **Stability Invariant:** Debounced highlight updates, graceful null-root handling, and recycled node safety.

ASCENDED MASTERY (VICTORY TIER)

- **On-Device GenAI:** On-device summarization via ML Kit / GenAI with status badge, warm-up, throttling, and quota management.
- **Prompt Customization:** Custom Prompt API, pinned floating palette, multi-window split support, and TalkBack-compatible output.
- **Sovereign Privacy:** Strictly offline local reasoning; optional cloud fallback behind an explicit off-by-default switch.



DELIVERABLES

- Public GitHub Repository with clean commit history
- README: Accessibility node traversal architecture, security model & failure modes
- Tool disclosure & privacy consent documentation

TECHNICAL FREEDOM

Construct using Native Android (Kotlin). Strictly read-only observer; zero unauthorized touch injection or task automation. Bonus marks for zero UI lag, low node traversal latency, and strict privacy safeguards.


Open

✦ PROBLEM STATEMENT // 01 ✦

THE MYSTIC FORGE

"No ancient grimoire dictates your path. Propose and forge any original artifact that lives natively on the phone—bound only by utility, audacity, and flawless craft."

I. THE INCEPTION COVENANT

- **Authentic Installation:** Must be a real installed application—not a generic website in a browser wrapper.
- **Abstract Scroll:** Submit a crisp architectural summary declaring problem, intended mortals, and why mobile is necessary.
- **Zero Video Mandate:** Defend live in viva; no prerecorded video tapes required.

II. FOUNDATIONAL SCOPE (PASS)

- **Operational Core:** At least 1 complete, polished workflow functional immediately upon launch.
- **Self-Contained Run:** Zero fragile dependencies; runs cleanly out of the box.
- **Git Commit Provenance:** Clear, incremental commits documenting true architectural evolution.

III. ASCENDED MASTERY (VICTORY)

- **Native Capabilities:** Offline operation, background watchers, widgets, hardware sensor telemetry, or system tray.
- **Measurable Telemetry:** Concrete numbers shown—bytes saved, search latency (ms), or memory footprint.
- **Failure Modes:** README analysis of 3 edge-case breakdowns and recovery paths.

IV. DEFENSE & EVALUATION

- **The Viva Crucible:** Explain your hardest technical decision directly before the Council.
- **Tool Disclosure:** State every library, framework, and AI accelerator in your README.
- **Code Rigor:** Clean, maintainable structure with zero unverified black-box code.

TECHNICAL FREEDOM OF THE NEXUS

Forge using any technology stack: Kotlin, Jetpack Compose, Flutter, Tauri, Electron, Qt, or Python. Offline core and local resilience earn highest honors.

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EVENT COORDINATORS

For any discrepancies, queries, technical doubts, or help during the event, reach out to the event coordinators below or any Computer Club (CC) member.

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IMPORTANT GUIDELINES & SUPPORT

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Organized by **Computer Club (CC), MNNIT Allahabad**.